Research Statement

I am an applied macroeconomist with a focus on development economics and the economics of education. My research so far has focused on the economics of education particularly for disadvantaged students in developing countries. Lack of education because of early dropout, or dysfunctional learning environments is a key constraint keeping countries poor. I design and test practical solutions to problems that bottleneck learning or keep students out of school. I also develop theory to explain how students form foundational reading skills crucial to becoming full participants in a modern economy.

Learning is a complex beast that depends on the conflicting interests of many stakeholders. While we have discovered many effective ways of improving learning by targeting teachers, administrators, or parents, all these stakeholders will need to come together to close the staggeringly large reading gaps between rich and poor countries; education systems in rich countries already run this way. Unexpected interactions between these programs can cause "voltage drop" and prevent programs from scaling effectively. Parents, teachers, or administrators can and do adjust their effort in response to schooling programs which is a key, poorly understood reason why these seemingly successful programs can fail when brought to hundreds of schools.

My job market paper furthers this research agenda. I cross-randomize two interventions: 1) organized parent meetings that give parents tools to understand how literate their child is and encourage reading at home; 2) structured management training of principals on how to coach their teachers to more effectively teach literacy and interact with parents. The key mechanism through which each program works is through unexpected interactions with inputs to the interventions. In particular, the management training for principals decreased reading ability for treated students because the additional coaching pushed teachers to improvise more during their lessons, which reduced their ability to adapt to the needs of their

students. By cross-randomizing the interventions, I further study the issue of stakeholder interactions by seeing how parents, teachers, and students respond when they receive *both* interventions. Parents (or siblings) increase their effort supporting students at home, and students practice reading more frequently. This increase in effort was enough to almost fully offset the decrease in reading ability from the principal intervention alone. This is a key mechanism for why some education programs succeed and why some fail, particularly at scale.

I am particularly interested in studying exactly how learning occurs. This goes beyond simple impact evaluations of education programs asking if the average student improved to uncover who the important stakeholders are in producing learning (students, parents, teachers, administrators) and what mechanisms drive successful programs. I do this by combining randomized field experiments with cutting-edge statistical and econometric techniques and economic theory. For example, in my study "How to Build a Reader: Evidence From a Scalable Literacy Intervention in Ghana, my co-authors and I tested a large-scale education program that combined structured pedagogy and targeted instruction. Structured pedagogy upskills teachers by providing them with comprehensive lesson guides so they can spend less time planning lessons and more time interacting with their students. Targeted instruction (sometimes called teaching at the right level) complements this by designing classrooms around children's skill rather than their age which allows teachers to adapt lessons to spend time on individual students rather than the average student. These are the two most effective programs so far developed for improving reading.

We show that the program increased literacy among first-grade students in rural Ghana by a huge amount over the control group. Using economic theory, we were able to learn even more from the data about the mechanisms of the program: I developed a model of interlocking skill formation and showed it is consistent with the data. The model let us understand which students were most impacted by the program, and how the program could best target

instruction at specific literacy skills and in what order. I built a machine learning, causal forest model with which we show that the program benefits nearly all students, no matter what language they speak at home or how wealthy their family is. We are continuing to study this program as it scales. Currently we are running versions of it in 180 government schools, which will eventually expand to schools covering entire regions of Ghana.

Materials and effort are not the only bottlenecks to learning. In a related line of work, I am working with a separate team of co-authors on a project in North-West Nigeria to study the effects of a program designed to prevent gender-based violence in schools. Gender-based violence in schools can cause students to dropout early from school or demotivate them from learning. Our preliminary research shows that the program raises awareness of this sensitive issue among teachers and students, decreases the prevalence of gender-based violence, and may raise literacy levels. Following these promising results, we are expanding the program in an RCT covering over 350 schools to test if it can still work at scale.

My research is not confined to Africa nor to elementary school students. I have an ongoing study at a medium-sized university in California working to understand how to mitigate the negative effects of academic probation. The particular challenge in this setting is the university's lack of resources to allocate to proven methods to help guide students to academic recovery, such as heavy-touch counseling sessions, so we redesigned the administrative letter sent to students when they are put on probation using softer language and highlighting concrete strategies they can use to get off probation. We are in the process of tracking a cohort of students who received our modified letter over the course of multiple years to see if they drop out less than their peers. This research studies administrative barriers to learning faced by students.